French National Assembly Speech Dataset
Technical Report

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June 2026

Abstract

This report documents the construction of a unified French National Assembly speech dataset covering debates from 1998 to 2026. The final dataset combines a historical ParisParl corpus derived from official PDF debate reports with recent DILA open-data XML records. It also relies on structured deputy, mandate, party, and presidency reference tables in order to bind speeches to stable political entities. The target output is an upload-ready speech table with one row per speech and the minimal schema expected by the downstream Policorp / Kamil structure.

Contents

1 Objective

The objective of the project is to produce a clean, documented, and reproducible dataset of French National Assembly speeches. The final speech table must cover the period from 1998 to 2026, keep only speech-level rows, and expose a compact schema suitable for downstream analysis and database import:

Column	Description
`speech_id`	Sequential speech identifier after chronological sorting.
`speech_text`	Cleaned speech text.
`speech_interjection`	Attached parliamentary reaction or interjection, when available.
`speech_date`	Speech date in `DD-MM-YYYY` format.
`speaker_id`	Deputy identifier or presidency identifier.
`country_id`	Country reference, fixed to `3` for France.
`term`	French legislature number at the speech date.

The final dataframe is named `df_1998_2026` in the exploratory notebook and is exported to `data/final/speeches_1998_2026.csv`.

2 Dataset Inventory

The local repository contains raw sources, intermediate transformed files, and final analysis-ready tables. Table 1 reports the principal CSV outputs documented in `dataset.md`. Counts are based on local file line counts, excluding the header row.

Local file	Records	Role
`data/final/speeches_1998_2026.csv`	5,557,923	Final unified speech table.
`data/final/parisparl_speeches_1998_2010_processed.csv`	1,424,597	Historical ParisParl input for the final merge.
`data/final/parisparl_speeches_1998_2019_processed.csv`	2,386,200	Extended processed ParisParl output.
`data/speech/2011_2026/centralized_speeches.csv`	3,084,895	Centralized DILA XML-derived speech table.
`data/final/politician.csv`	3,645	Deputy reference table.
`data/final/mandate.csv`	5,193	Mandate reference table.
`data/parties/party.csv`	41	Party reference table.
`data/final/presidency.csv`	9	Presidency reference table.

Table 1: Main local dataset files.

3 Primary Speech Sources

3.1 DILA Open Data, 2011–2026

The recent speech corpus is built from the DILA open-data archive: `https://echanges.dila.gouv.fr/OPENDATA/Debats/AN/`. DILA is the French government directorate responsible for legal and administrative information. For recent French National Assembly debates, this is the preferred source because it provides the original parliamentary XML structure published by the administration.

The local DILA pipeline has three reproducible stages:

1. **Download and extraction.** `converter/download_dila_debats.py` reads the DILA index, detects yearly archives, downloads the requested years, and extracts XML files into `data/speech/2011_2026/raw`. The extraction keeps the year/session directory structure and protects against unsafe archive paths.

2. **XML conversion.** `converter/convert.py` parses each `CRI_*.xml` debate file and writes structured JSON documents under `data/speech/2011_2026/converted`. The parser preserves the paired `AAA_*.xml` source name when present and extracts session metadata, paragraph identifiers, speaker blocks, speaker links, speaker roles, and the nested debate hierarchy.

3. **CSV centralization.** `converter/json_to_csv.py` flattens converted JSON files into `data/speech/2011_2026/centralized_speeches.csv`. The result contains one row per paragraph and includes raw text, cleaned text, topic hierarchy, speaker metadata, interjections, quotes, and session numbers.

The DILA cleaning stage is required because the XML source is parliamentary markup rather than an analysis-ready table. Paragraphs may include nested speaker tags, non-breaking spaces, repeated whitespace, procedural titles, interruptions, and section headers mixed with spoken content. The converter therefore stores both raw paragraph text and normalized paragraph text, while separating speaker metadata from the speech content.

3.2 ParisParl, 1998–2010 for the Final Merge

The historical speech corpus is based on the ParisParl corpus: `https://zenodo.org/records/3819374`. ParisParl was produced in the PolMine project from official PDF reports of plenary sessions in the French National Assembly. It is used to extend the speech coverage before the DILA XML period.

The original tabular version was no longer available, but the encoded Corpus Workbench version was recovered and reversed into yearly CSV files stored under:

`data/speech/1996_2019/parisparl_reden_YYYY.csv`

Although the source description covers 1996 to 2019, the local processed files currently start in 1998. The first two years are documented as missing in the data quality review.

Cleaning ParisParl is more complex than cleaning DILA because the source is PDF-derived rather than native structured XML. The extracted text stream mixes OCR and detokenization artifacts, page-layout noise, table-of-contents lines, procedural headings, major topics, subtopics, speeches, and interjections. Numeric strings can represent article numbers or page references rather than topic numbers, and presidency or procedure rows can be incorrectly interpreted as speeches if they are not separated.

The main ParisParl restructuring components are:

- `mine_parisparl.py` and `eda/mine_parisparl.ipynb`, used to prototype and apply structure recovery rules.

- `eda/summary_extraction.py`, which extracts the official `SOMMAIRE` zone and uses it as the preferred source for topic order, major topics, and subtopics.

- `eda/title_cleaning.py`, which fixes French detokenization artifacts in parliamentary titles.

- eda/interjection_filter.py, which keeps genuine reactions such as applause, laughter, protests, and interruptions while filtering structural false positives.
- eda/speech_grouping.py, which groups consecutive rows from the same speaker, keeps procedure rows separate, and attaches reaction interjections to the relevant speech group.
- eda/entity_binding.py, which links cleaned speeches to party and politician references.

4 Reference Entity Tables

4.1 Deputies and Mandates

The deputy and mandate tables provide stable political entities for both speech corpora. Speeches need deputy identifiers, mandate periods, legislature numbers, and party or group information so that each row can be linked to a person, a parliamentary term, and a political family.

The primary structured source is the official Assemblee nationale historical deputy archive: https://data.assemblee-nationale.fr/acteurs/historique-des-deputes. Local JSON files are stored in:

- data/deputes/acteur
- data/deputes/organe
- data/deputes/deport

The conversion is implemented in convert_acteurs_to_csv.py. It extracts one politician row per PA... identifier and one mandate row per mp_id + legislature. The script also extracts birth information, profession, institutional status, constituency, mandate dates, election dates, parliamentary groups, committee memberships, source URLs, and a country reference.

The Sycomore website is used as a complementary source for legislatures 10 to 17: https://www2.assemblee-nationale.fr/sycomore/recherche. It fills missing politicians and mandates, especially for older legislatures where the local structured JSON data can be incomplete.

The merge rules are conservative:

- official local Assemblee nationale JSON rows take priority over Sycomore rows;
- politicians are deduplicated by mp_id;
- mandates are deduplicated by mp_id + term;
- existing PA... identifiers are preserved;
- generated fallback identifiers are recorded in audit reports;
- each mandate must point to an existing politician.

4.2 Political Parties

data/parties/party.csv is a manually curated reference table used to standardize party and parliamentary group labels across deputy, mandate, and speech datasets. It maps unstable historical labels, acronyms, group names, and aliases to stable internal party identifiers.

This table is used by convert_acteurs_to_csv.py when converting mandate group labels to party_id values and by eda/entity_binding.py when binding speech rows to parties. Labels are normalized before matching: case, accents, punctuation, parenthetical details, and known historical variants are handled.

Rows that cannot be mapped safely are not guessed. In the deputy pipeline, they are emitted as empty or P000 values and written to deputes_boostrap/report/unresolved_party_ids.csv for manual review.

4.3 Presidency

`data/final/presidency.csv` is used to bind speeches spoken by the chair of the Assembly when no deputy identifier is available. During the final merge, rows identified as presidency speeches are matched to a presidency identifier according to the speech date.

5 Final Merge for the 1998–2026 Speech Table

The final upload-ready speech table combines:

- `data/final/parisparl_speeches_1998_2010_processed.csv`, for historical ParisParl speeches from 1998 to 2010;
- `data/speech/2011_2026/centralized_speeches.csv`, for DILA open-data debates from 2011 to 2026;
- `data/final/presidency.csv`, for presidency identifier binding.

The merge logic follows these steps:

1. Reduce DILA rows to the columns needed by the final schema: date, cleaned text, deputy identifier, speaker label, and interjection.
2. Drop rows without a speaker, except presidency rows that can be bound later through the presidency table.
3. Rename DILA columns to the final naming convention.
4. Recover missing DILA dates from source file names when possible.
5. Read the processed ParisParl table and restrict it to 1998–2010.
6. Keep only ParisParl rows with `row_type == "speech"`.
7. Rename ParisParl columns to the final naming convention.
8. Concatenate the normalized ParisParl and DILA speech tables in chronological order while preserving source order within each corpus.
9. Fill missing presidency speaker identifiers from `presidency.csv` when the row is spoken by the Assembly presidency.
10. Set `country_id = 3` for France.
11. Compute `term` from the speech date using the French legislature date ranges from the 11th to the 17th legislature.
12. Generate `speech_id` after sorting, starting at 1.
13. Remove temporary role, parsed-date, and source-order columns.

6 Quality Assurance

The project includes targeted tests under `tests/` and notebook-level quality checks for the final merged table. The final notebook asserts that:

- final columns match the expected schema and order;
- `speech_id` is unique and monotonically increasing;
- `speech_date` is never missing;
- `speaker_id` is never empty after presidency binding;
- `country_id` is always `3`;
- `term` is never missing.

Rows that still have no speaker identifier after presidency binding are removed from the upload dataframe because the target structure requires a bound speaker.

The deputy pipeline also writes audit reports under `deputes_boostrap/report`, including new politicians, new mandates, fallback identifiers, duplicates, and unresolved party identifiers.

7 Known Limitations

- The ParisParl source is documented as covering 1996–2019, but the local processed files currently start in 1998. The first two years remain a known coverage gap.
- ParisParl is reconstructed from PDF-derived text. Even after cleaning, it is more exposed to OCR artifacts, layout noise, and false structural positives than the DILA XML source.
- Some fields in the politician target schema are intentionally empty because they are not available in the local source JSON files: `education_level`, `school`, `academic_title`, `languages`, `list`, and `votes`.
- Party labels are historically unstable. The curated party table reduces this problem, but unresolved labels are kept for manual review instead of being guessed automatically.
- The final table keeps the compact Kamil structure. Richer source metadata such as detailed topic hierarchy, raw paragraph identifiers, and source filenames remains available in intermediate files rather than in the final upload table.

8 Reproducibility

The DILA pipeline can be run with:

```
python3 main.py download --years 2011-2026
python3 main.py convert
python3 main.py csv
```

The full DILA pipeline can also be run with:

```
python3 main.py all --years 2011-2026
```

The deputy and mandate pipeline can be run with:

```
python3 convert_acteurs_to_csv.py
```